

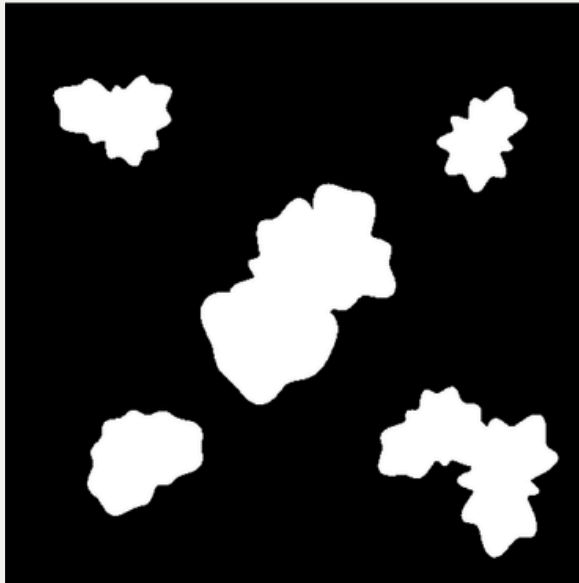


Tree Crown Instance Segmentation

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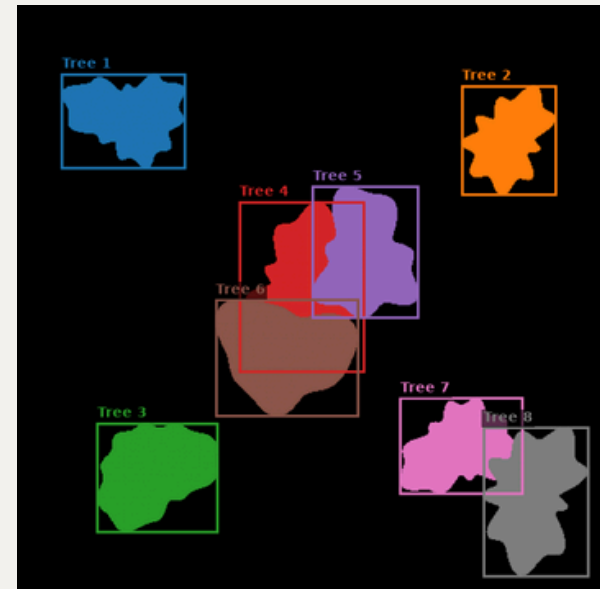
Why Tree Instance Segmentation?

semantic
segmentation



limited for tree-level analysis

instance
segmentation

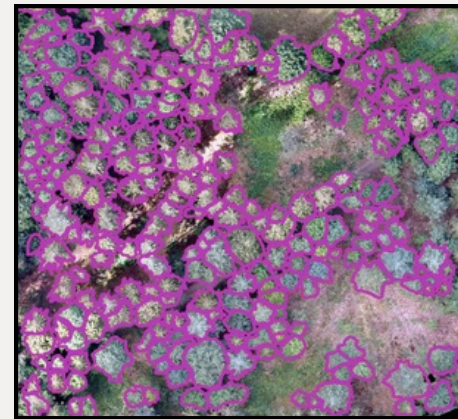
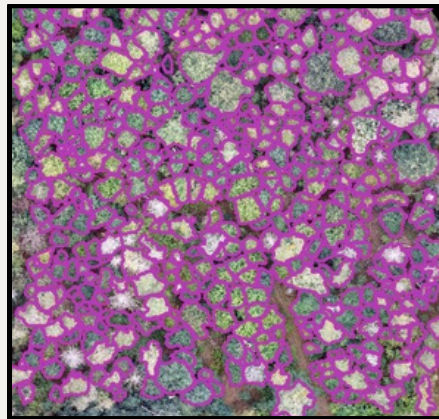


useful for counting, crown
metrics, tree level species
classification ...

Training Data

BamForests [1]:

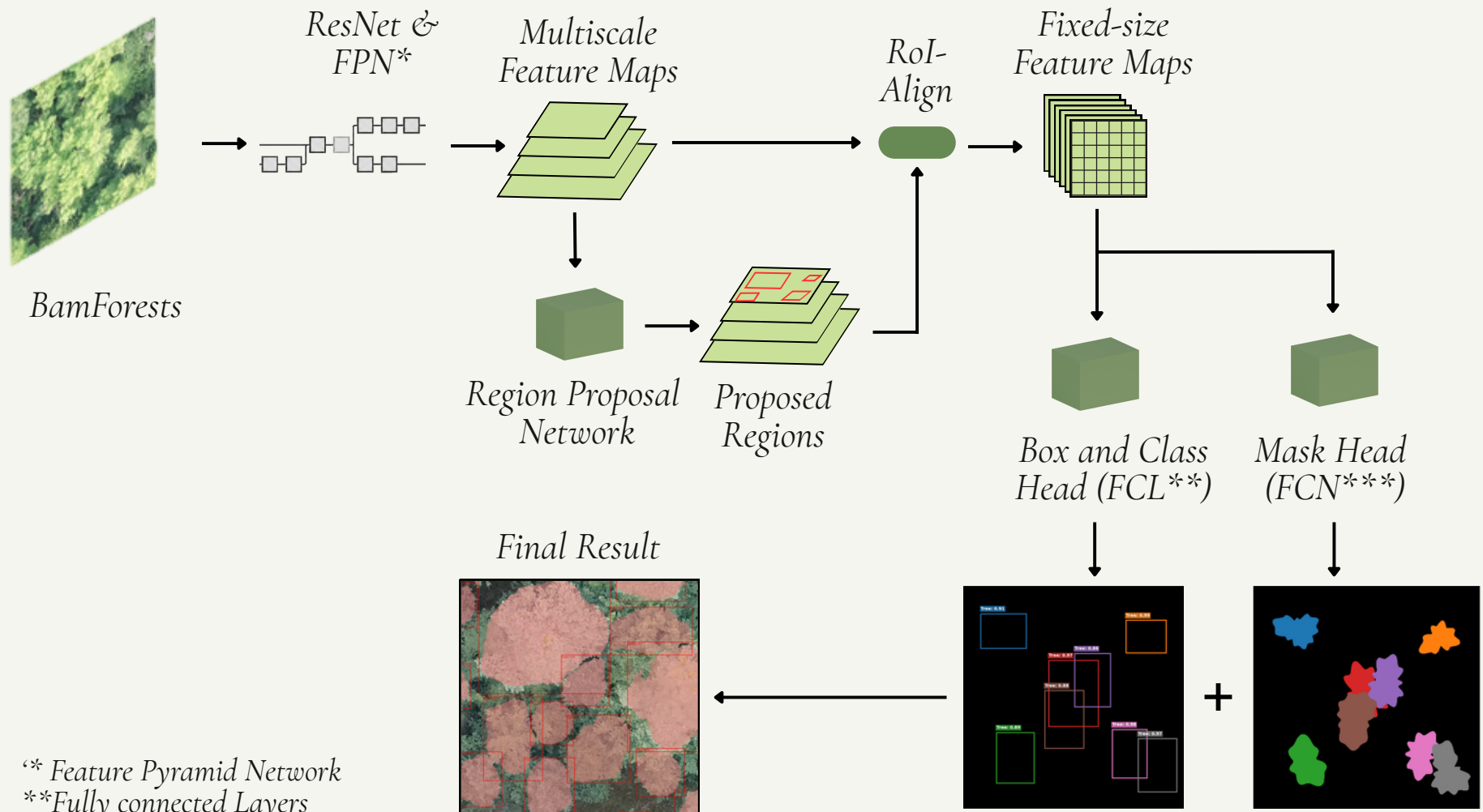
- ready-to-use labeled RGB-dataset (COCO-format) with 27.160 trees in total (split in train/test/validation)
- 4 AOIs near Bamberg (Upper Franconia)



Mask R-CNN

- extension of Fast R-CNN for instance segmentation [2]
- added a branch for object mask to Faster R-CNN
- often used for tree crown segmentation [3,4,5]

Model architecture



*** Feature Pyramid Network*

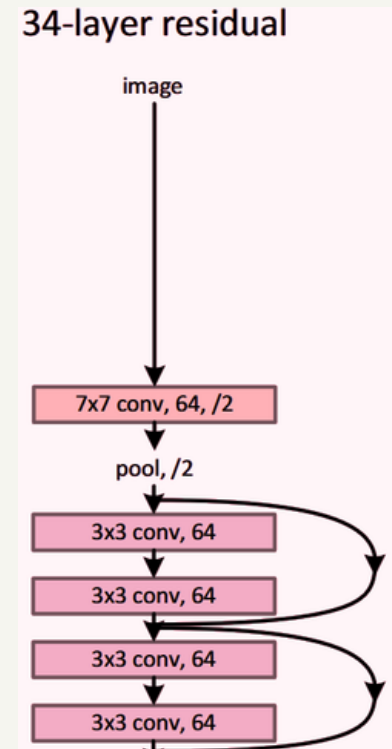
***Fully connected Layers*

**** Full convolutional Network*

Backbone

ResNet34 [6]

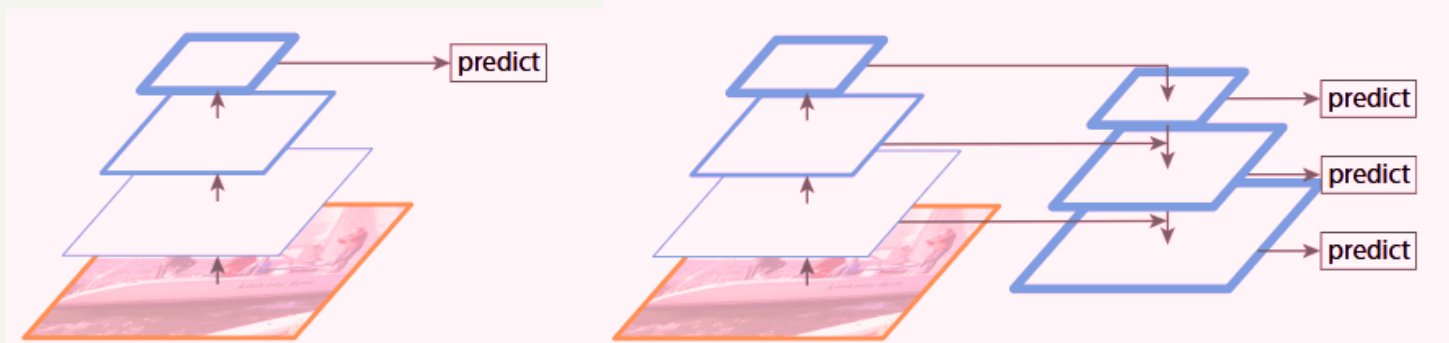
- Deep NN with skip connections
- pretrained weights were downloaded



Backbone

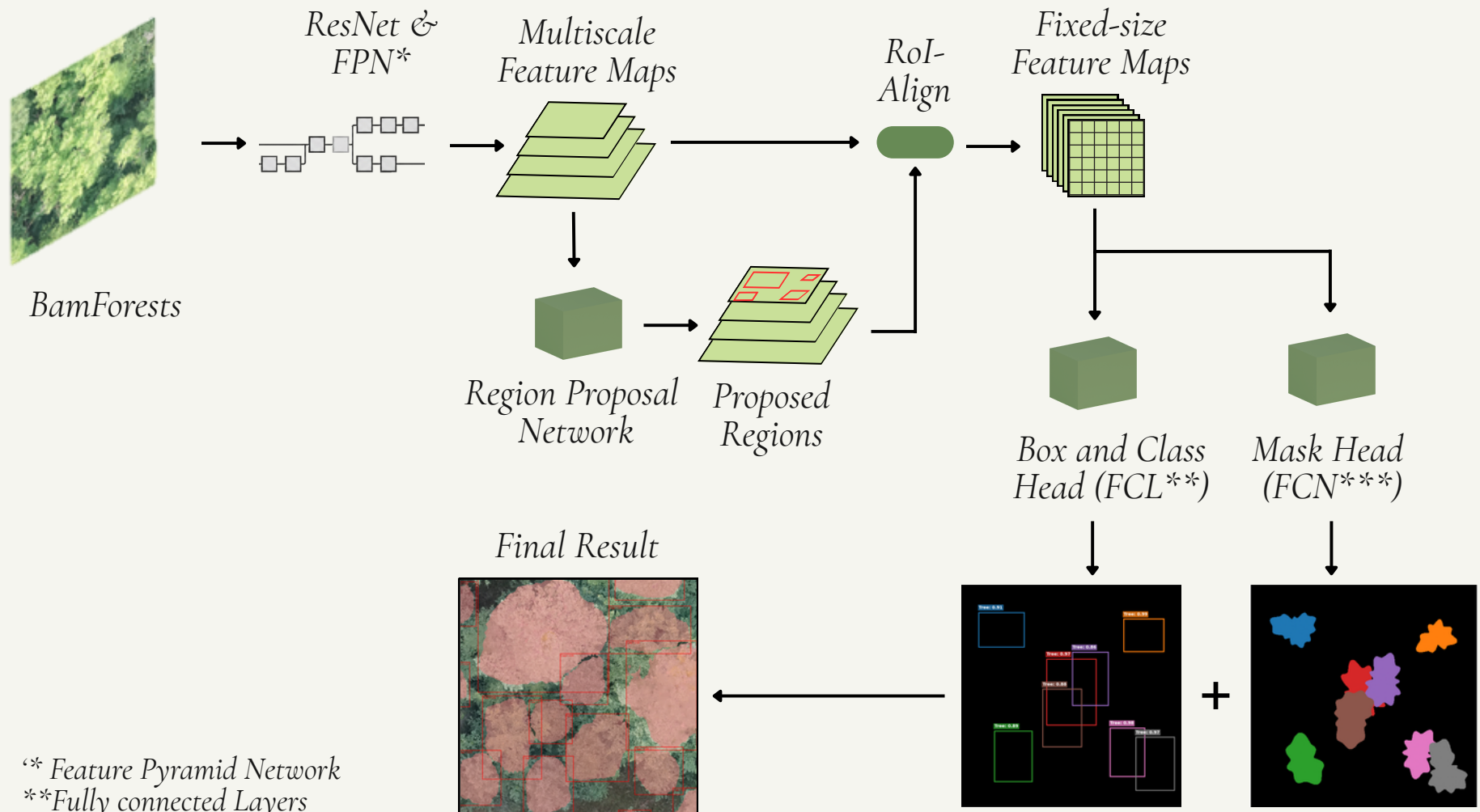
Feature Pyramid Network (FPN) [7]

- FPN create multiscale feature maps



Single feature map vs. FPN [7]

Model architecture



* Feature Pyramid Network
 ** Fully connected Layers
 *** Full convolutional Network

AdamW-Optimizer [8]

Adaptive Moment Estimation with Weight Decay

momentum

moving average of the gradients:

→ *controls direction and smooths noise to overcome local minima*

adaptive scaling

moving average of the squared gradients:

→ *adjusts the step size for each parameter*

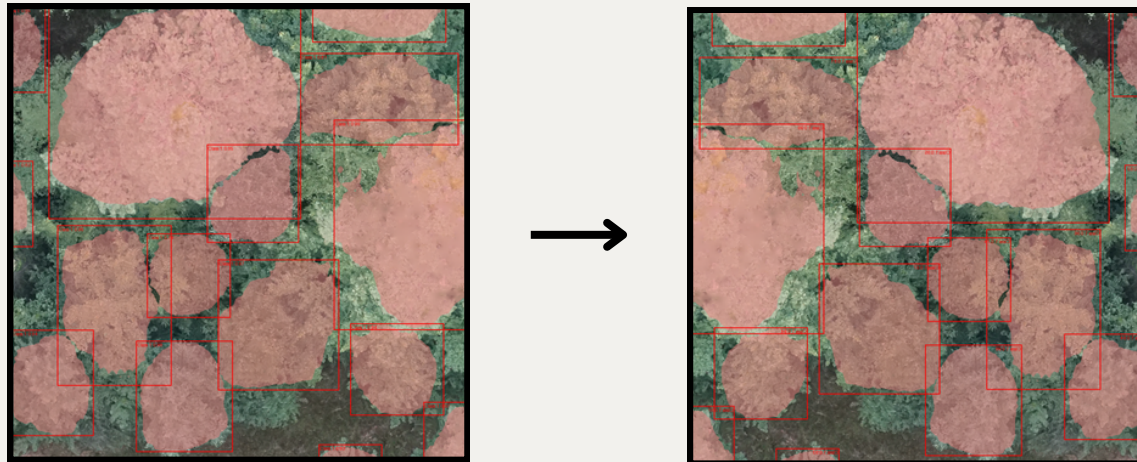
Decoupled weight decay

separates weight shrinkage directly from the gradient update step:

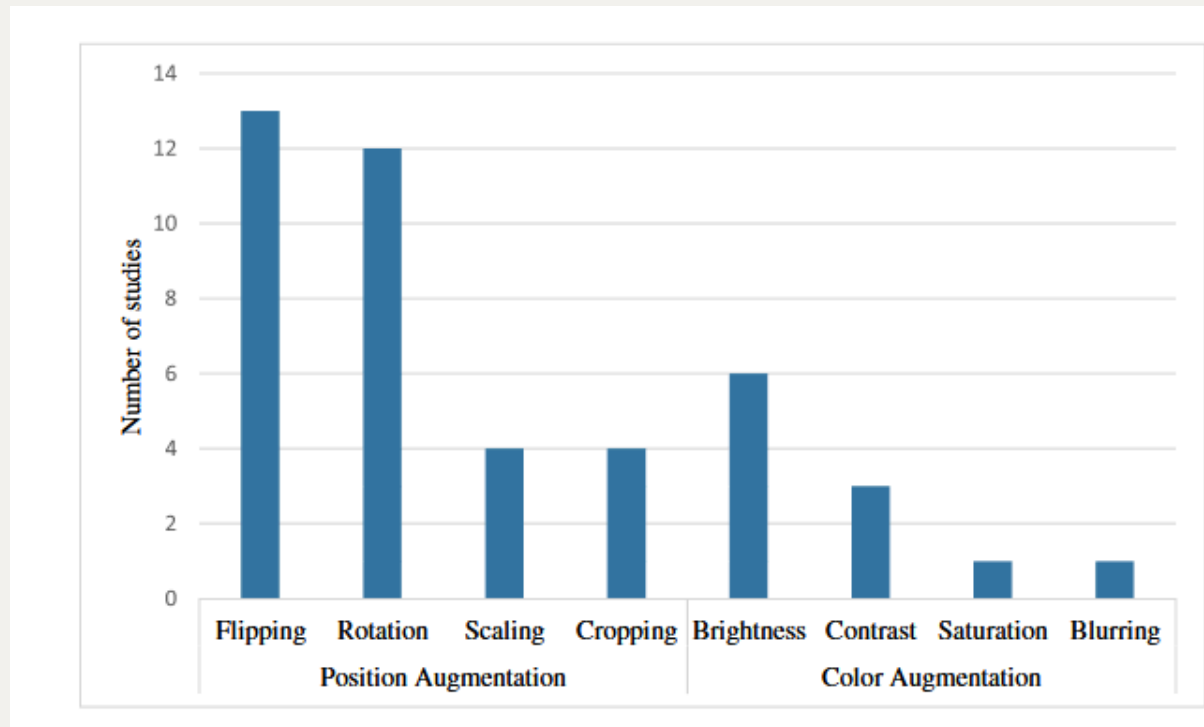
→ *L2 regularization*

Augmentation

- Rotation
- Reflection
- Scaling
- Brightness and Contrast
- (slight) Gaussian Blur

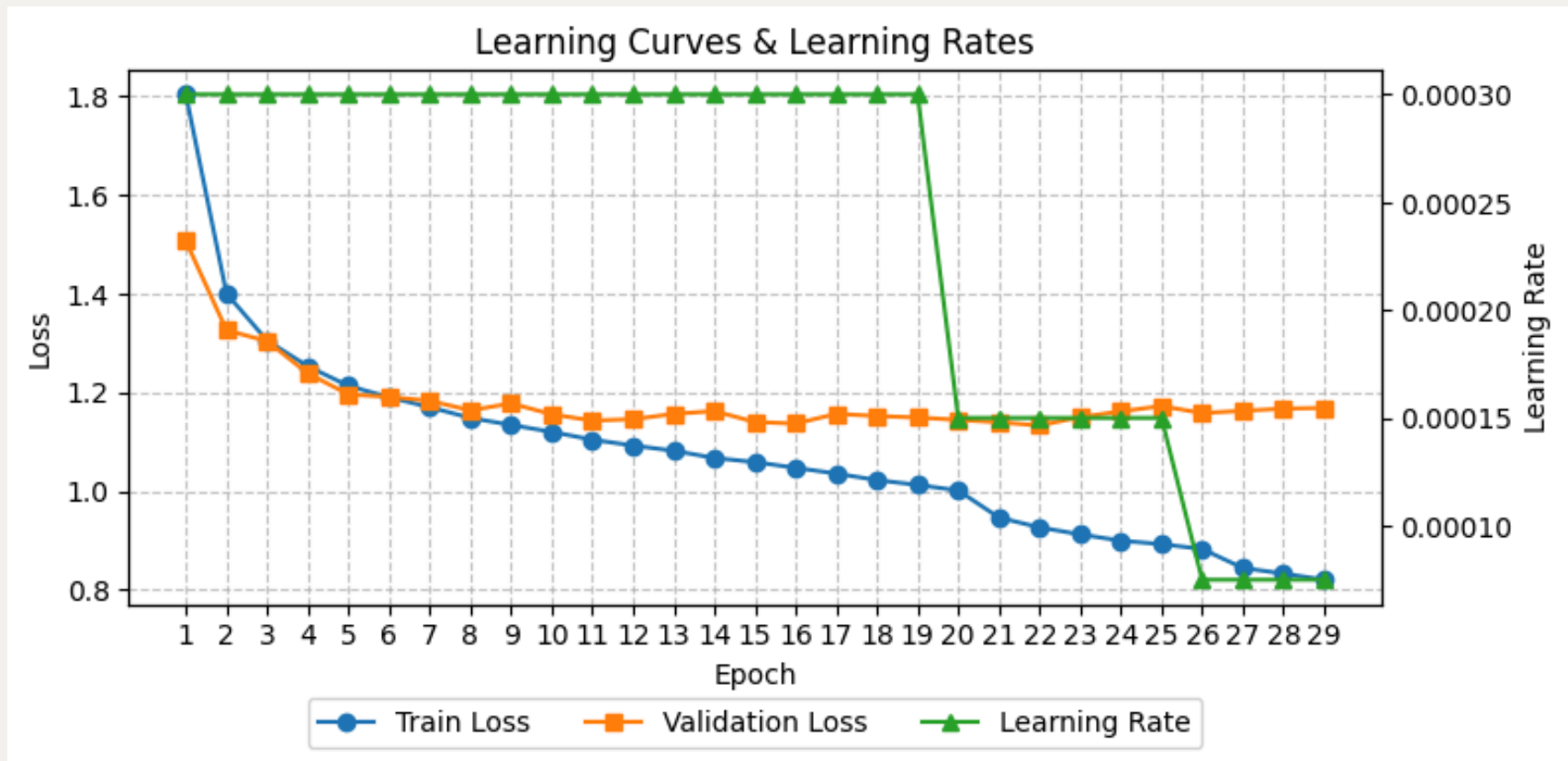


Augmentation



Most used augmentation techniques for crown segmentation according to [9]

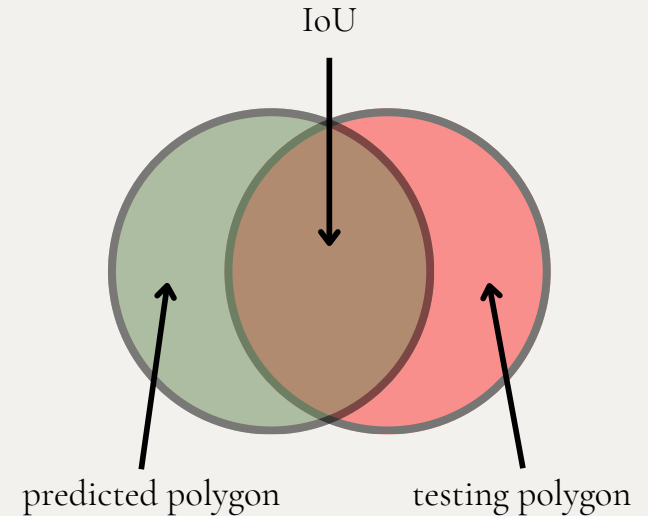
Learning Curves



Accuracy Assessment

- Intersection over Union (IoU)

$$\text{IoU}(A,B) = \frac{A \cap B}{A \cup B}$$



→ **IoU** > **0.5** is typically regarded as a **true positive** [4,5]

Therefore, classic parameters as **Precision**, **Recall** and **F1-Scores** can be calculated.

Accuracy Assessment

My model:

Accuracy: 0.525

F1 Score: 0.688

Precision: 0.637

Recall: 0.749

→ False Positives > False Negatives

Accuracy Assessment

My model:

Accuracy: 0.525

F1 Score: 0.688

Precision: 0.637

Recall: 0.749

≈

**Reference model
[4]:**

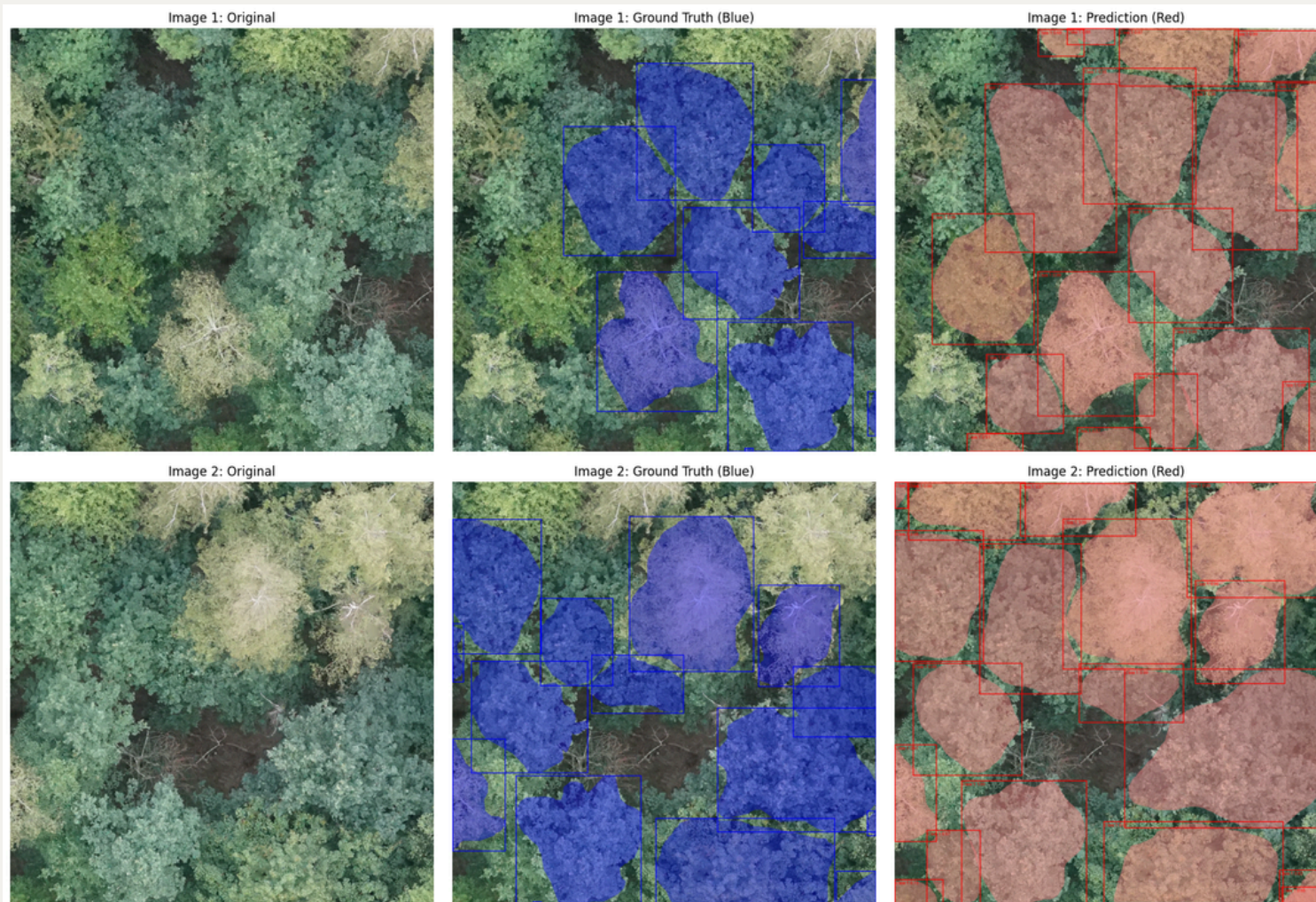
F1 Score: 0.634

Precision: 0.640

Recall: 0.629

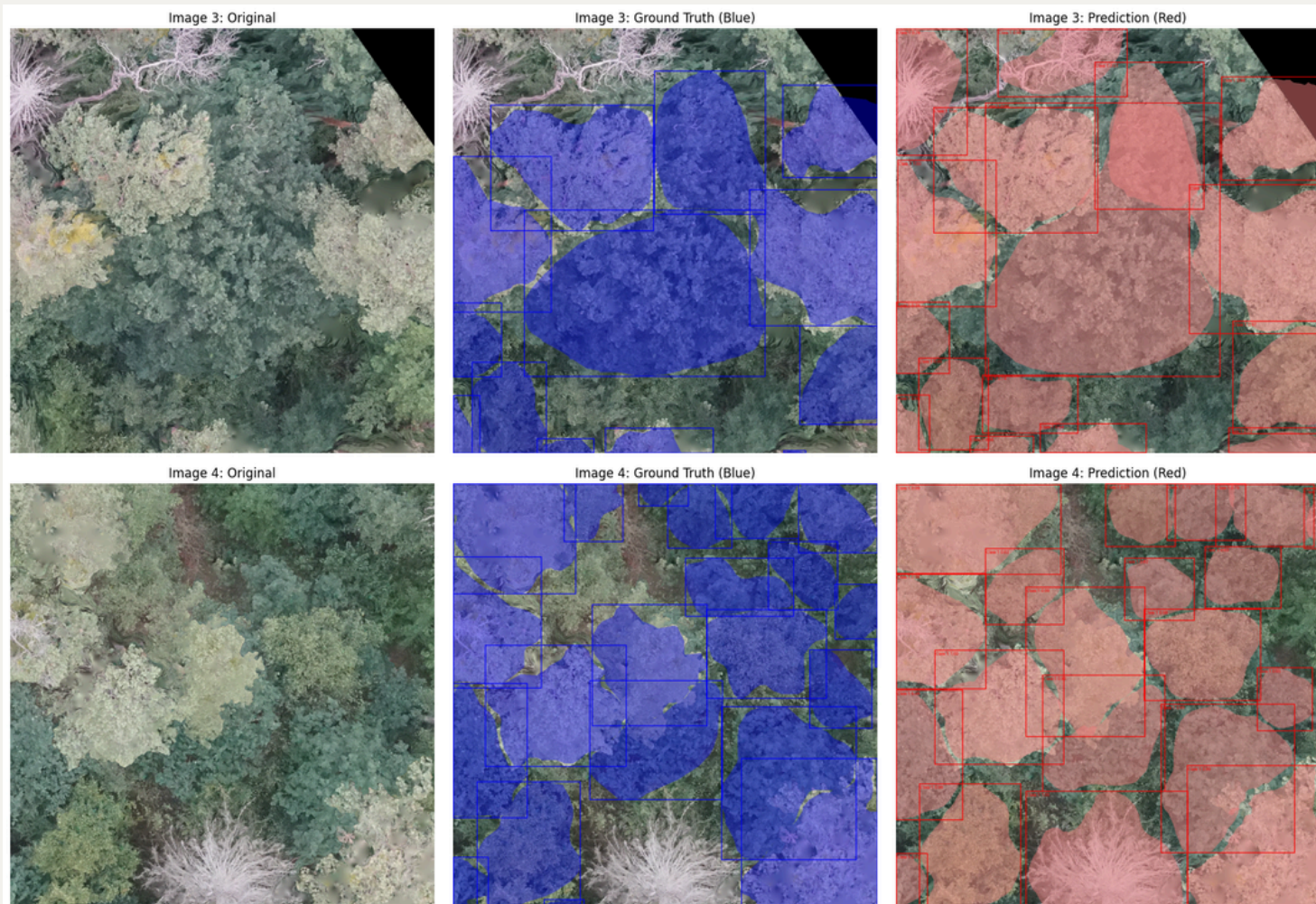
Examples

Randomly selected



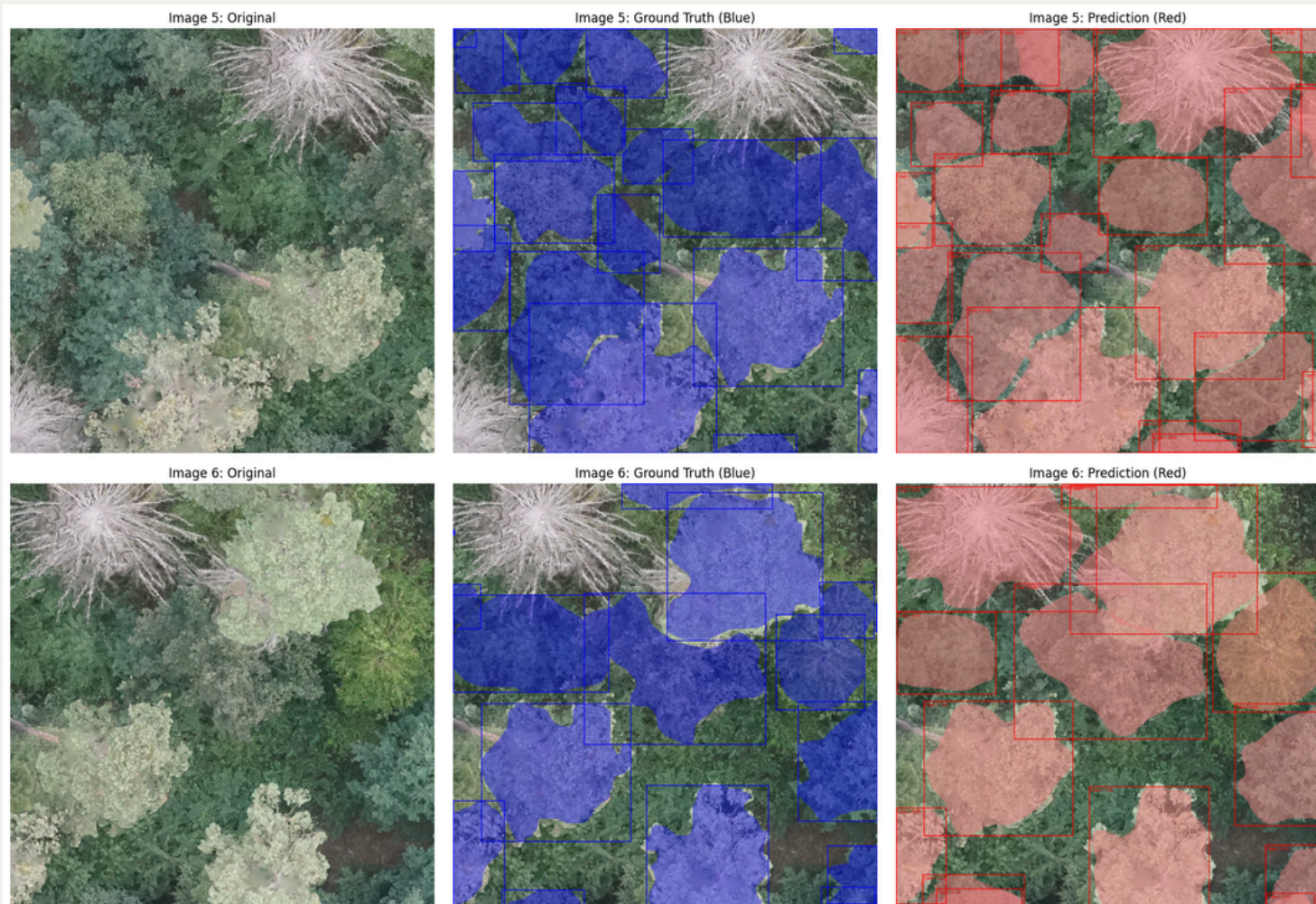
Examples

Randomly selected



Examples

Randomly selected



Sources

1. Troles J, Schmid U, Fan W, Tian J (2024)

BAMFORESTS: Bamberg Benchmark Forest Dataset of Individual Tree Crowns in Very-High-Resolution UAV Images. Remote Sensing, 16(11):1935.

2. He K, Gkioxari G, Dollár P, Girshick R (2017)

Mask R-CNN. arXiv:1703.06870.

3. Hao Z, Lin L, Post CJ, Mikhailova EA, Li M, Chen Y, et al. (2021)

Automated tree-crown and height detection in a young forest plantation using mask region-based convolutional neural network (Mask R-CNN)

ISPRS Journal of Photogrammetry and Remote Sensing, 178:112–123.

4. Ball JGC, Hickman SHM, Jackson TD, Koay XJ, Hirst J, Jay W, et al. (2023)

Accurate delineation of individual tree crowns in tropical forests from aerial RGB imagery using Mask R-CNN. Remote Sensing in Ecology and Conservation, 9(5):641–655

5. Braga JRG, Peripato V, Dalagnol R, Ferreira MP, Tarabalka Y, Aragão LEOC, et al. (2020)

Tree Crown Delineation Algorithm Based on a Convolutional Neural Network. Remote Sensing, 12(8):1288.

Sources

6.He K, Zhang X, Ren S, Sun J (2016)

Deep Residual Learning for Image Recognition. CVPR, 770–778.

7.Lin TY, Dollár P, Girshick R, He K, Hariharan B, Belongie S (2017)

Feature Pyramid Networks for Object Detection. CVPR, 2117–2125.

8.Loshchilov I, Hutter F (2019)

Decoupled Weight Decay Regularization. ICLR.

9.Zhao H, Morgenroth J, Pearse G, Schindler J (2023)

A Systematic Review of Individual Tree Crown Detection and Delineation with Convolutional Neural Networks (CNN). Current Forestry Reports, 9:149–170.

Model architecture

